

P. BHUVANESH

Robotics Engineer | ROS 2 Simulation & Autonomous Robotics

Thiruvallur, Tamil Nadu | +91 75300 47857 | bhuvaneshash@gmail.com
GitHub: github.com/Bhu470/dashboard | Portfolio: bhu470.github.io/dashboard/

PROFESSIONAL SUMMARY

Robotics Engineer focused on ROS 2-based robot simulation, autonomous systems, and robotic mechanism development. Hands-on experience with Gazebo, RViz2, URDF/XACRO, TF2, SLAM, mapping, localization, Nav2 workflows, drone controller simulation, OpenCV, gait control, and inverse kinematics. Experienced in developing and validating mobile, legged, and aerial robot simulations in Linux/Ubuntu environments.

EXPERIENCE

Robotics Engineer | Evolve Robotics Lab, Chennai | Mar 2025 – Present

- Developing ROS 2 robotic simulation modules, URDF/XACRO models, Gazebo environments, and RViz2 visualization workflows.
- Working on autonomous navigation, LiDAR/sensor simulation, TF validation, SLAM/mapping, localization, and robot-control workflows.
- Developing simulation concepts for AMR, spider/legged robots, and drone control.

Robotics Engineer Intern | Evolve Robotics Lab, Chennai | Oct 2024 – Mar 2025

- Developed ROS 2 nodes, Gazebo simulation workflows, URDF models, robot visualization, SLAM testing, and obstacle-avoidance workflows.

TECHNICAL SKILLS

ROS 2 & Simulation: ROS 2 Humble/Jazzy, Gazebo Sim, RViz2, URDF/XACRO, TF2, ROS 2 Launch, ros_gz, robot_state_publisher, joint_state_publisher, ros2_control

Autonomous Robotics: SLAM Toolbox, 2D LiDAR mapping, localization, Nav2, costmaps, path planning, obstacle avoidance, AMR, differential-drive robots

Robot Control & Mechanisms: Drone controller simulation, keyboard-based drone control, gait control, inverse kinematics (IK), multi-joint/legged robot control

Computer Vision: OpenCV, vision-based processing and prototyping

Programming & Platforms: Python, C++, Embedded C, Linux/Ubuntu, Git/GitHub

Embedded & Hardware: ESP32, Arduino, STM32, encoders, ultrasonic sensors, DC/servo motors, L298N, PCA9685

Communication: UART, SPI, I²C, PWM

PROJECTS

Drone Simulation – Design & Keyboard Control

- Designed a drone simulation workflow in Gazebo/ROS 2 and implemented keyboard-based control for simulation testing.

Spider Robot – Gait Control & Inverse Kinematics

- Developed a multi-leg robot simulation using URDF/XACRO with gait-control concepts and inverse-kinematics-based leg movement.

ROS 2 AMR – Mapping, Localization & Navigation

- Developed an autonomous mobile robot simulation using URDF/XACRO, Gazebo, RViz2, LiDAR, SLAM/mapping, localization, and Nav2 workflows.

ROS 2 Mobile Robot Simulation

- Built and validated differential-drive simulation with TF inspection, encoder integration, sensor data, and autonomous-navigation workflows.

EDUCATION

B.E. – Electrical and Electronics Engineering | Sriram Engineering College | 2018 – 2022

Diploma – Electrical and Electronics Engineering | Sriram Polytechnic College | 2016 – 2018

CERTIFICATIONS

Robotics Internship – Evolve Robotics Lab | Embedded Systems – BYTESINBITS | MSP430 Programming – Udemy | IoT Training – BSNL